Pricing and Hedging Through a Benchmark Transition Without an Options Market

Identifiability, Smile Transport and Finite-Sample Model-Risk Certification for the JIBAR-to-ZARONIA Derivatives Book

by

Thabang Bongani Junior Baloyi

Student number 1113941

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School of Statistics and Actuarial Science

Faculty of Science

University of the Witwatersrand, Johannesburg



Supervisor: To be confirmed

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A concept note for discussion. Results described here are proposed targets, not proved.

The research problem

When an interest-rate benchmark is replaced, the cashflows of existing derivatives change: a forward-looking term rate fixed at the start of an accrual period gives way to a rate compounded over that period from overnight fixings. This alters both when the underlying is observed and how uncertainty accumulates, for contracts written before the replacement was contemplated.

The difficulty I wish to study is informational. A dealer may observe a reliable market in instruments referencing the legacy benchmark (curves, swaps, basis instruments and legacy options), while holding little or no direct evidence about options on the successor rate. Those observations constrain some features of the transition; they need not determine the full distribution that an option payoff depends on. An option is sensitive to outcomes beyond an average rate: two models can agree on forward values while assigning different probabilities to large moves, and a caplet distinguishes them. Importing a volatility parameter from the legacy market therefore yields a number without showing that it follows uniquely from what was observed.

A second and separate question concerns trading. Valuation uncertainty is the range of model prices compatible with the observations; trading risk is the realised profit or loss from holding and hedging the liability. A closely calibrated model can still produce large losses, and a small average hedging error does not imply a valid statement about the upper tail.

The South African JIBAR-to-ZARONIA transition supplies the application, with South African Reserve Bank conventions fixing the contractual implementation. Any claim of illiquidity will be dated and tied to a specific instrument, tenor and observation period, supported by a data audit rather than assumed.

Central question and research questions

> Under what assumptions can legacy market information determine successor-rate option values, and what statistically defensible bounds can be placed on the losses from hedging those options during a benchmark transition?

- **RQ1.** Which successor valuation functionals are constant across the models compatible with legacy option quotes and curve or basis observations?
- **RQ2.** Which additional observations or restrictions restore identification, and how stable is that conclusion to quote noise?
- **RQ3.** Under what dependence and distribution-shift assumptions can a calibrated loss threshold retain a finite-sample guarantee?
- **RQ4.** Are the resulting price ranges and risk bounds informative for a realistic hedge, once costs and data limitations are included?

Aim and objectives

Aim. To develop and evaluate a framework for identifying successor-rate option valuations and quantifying residual hedging risk under limited option-market information.

1. Specify a coherent transition model, contractual conventions and observable market information, keeping risk-neutral pricing assumptions separate from historical estimation.
2. Prove identification or non-identification results for a restricted model class, and determine the resulting valuation ranges or approximation errors.
3. Construct implementable hedges from available instruments and investigate finite-sample loss guarantees under explicit assumptions.
4. Evaluate pricing uncertainty, hedge performance and certificate informativeness in simulation and in an appropriately limited empirical application.

Proposed approach

The work begins with a low-dimensional Markov model carrying an overnight-rate factor and a spread factor under an explicit pricing measure. A two-factor Gaussian diffusion is the initial analytical laboratory: it admits explicit covariance calculations and tractable simulation, and its limitations, including the possibility of negative rates, will be recorded rather than concealed. The observation map from model to traded instrument prices will price actual cashflows before any identification claim is made.

The first analytical question is identification. Let O denote the observations available at a valuation date and $\Lambda(\theta)$ the model-implied observation vector. The compatible set is

$$\Theta(O) = \{\theta \in \Theta : \Lambda(\theta) \text{ is compatible with } O\}, \tag{1}$$

with compatibility defined by quote tolerances. A target valuation V_θ is identified if it is constant on $\Theta(O)$; otherwise the attainable range is

$$V_- = \inf_{\theta\in\Theta(O)} V_\theta, \qquad V_+ = \sup_{\theta\in\Theta(O)} V_\theta, \tag{2}$$

and these are identification ranges, not confidence intervals.

A one-period calculation shows the mechanism. Let a successor shock, a basis shock and the resulting legacy shock be

$$X = \sigma Z_1, \qquad B = \eta(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2), \qquad Y = X + B, \tag{3}$$

with Z_1, Z_2 independent standard normal. If an idealised legacy surface identifies Y as standard normal, that is one equation in the three covariance parameters. At $\rho = 0$ both $(\sigma, \eta) = (1/\sqrt{2},\, 1/\sqrt{2})$ and $(\sigma, \eta) = (1/2,\, \sqrt{3}/2)$ reproduce it, yet the zero-strike call on X is worth $\sigma/\sqrt{2\pi}$, that is 0.282 in the first case and 0.199 in the second.

Agreement on the full legacy distribution has not determined the successor payoff. The thesis must establish whether the required variation survives once curves, basis instruments, several maturities and any available options enter the observation map; parameter counting and a good numerical fit are not sufficient.

The hedging analysis measures realised risk. The signed hedging loss is the discounted liability less terminal hedging wealth, where that wealth is initial capital plus trading gains net of costs. A sensitivity hedge and a mean-variance hedge provide the initial comparisons, with admissibility, financing, rebalancing and costs fixed before evaluation and compared across compatible pricing models on common scenarios.

The statistical object is an upper threshold for a future hedging loss, not a claim that the price is correct. The baseline is split conformal calibration for a hedge frozen before calibration, whose order-statistic threshold has an exact finite-sample marginal guarantee under exchangeability. The research task is the extension, namely coupling under a stated mixing condition to control dependence and a distance between the calibration and test loss distributions to control shift, giving a target of the schematic form

$$\Pr\{L_{\text{new}} \le q\} \;\ge\; 1 - \alpha - \varepsilon_{\text{dep}} - \varepsilon_{\text{shift}} - \varepsilon_{\text{est}}. \tag{4}$$

This display is a research target, not a proved theorem. A bound that is vacuous will be reported as an inability to certify under the stated assumptions.

Scope, positioning and the provisional gap

The initial product is caplets; the initial model is a small factor system; the initial hedge set is liquid swaps and cash instruments, restricted to what the data audit shows was tradeable. A successor option will not be used as a hedge merely because it would complete the market. Swaptions, large portfolios, jumps, multi-tenor consistency, learned hedging policies and rough-path methods are treated as extensions, each requiring a specific reason tied to a research question.

Brigo and Mercurio (2006) supply the pricing architecture; Lyashenko and Mercurio (2019) treat backward-looking rates; Willems (2020) studies SABR smiles for risk-free-rate caplets. Local work supplies method and context: Konaite (2024), Menziwa (2023), Robbertze (2021) and Stangroom (2023), together with Mavuso (2014) on mean-variance hedging where a liquid asset is traded dynamically and an illiquid asset is held statically.

Alfeus (2026) models JIBAR-to-ZARONIA spread dynamics, which is historical spread modelling and is distinct from identifying a risk-neutral valuation. The statistical component builds on conformal prediction (Vovk 2012; Barber et al. 2023; Oliveira et al. 2024; Barber and Pananjady 2026; Ramos et al. 2026) rather than proposing a new coverage principle.

The provisional gap is the connection between a rigorously characterised set of transition models and a finite-sample analysis of the hedging losses they imply, under stated dependence and distributional change. The first phase is a structured comparison in common notation; if the candidate results reduce to existing theorems, the contribution will be narrowed rather than inflated.

Validation, feasibility and contribution

Three settings will be used. Controlled simulation generates data from specified transition models with known targets, including deliberately misspecified models. Retrospective validation, if licensed and comparable data can be obtained, withholds successor option quotes while fitting to permitted legacy and curve information and reveals them only for evaluation; a historical LIBOR-to-SOFR dataset is a candidate.

The South African application proceeds on the instruments and dates the data audit supports, reporting compatible valuation ranges and observable hedge outcomes where successor quotes are absent. Training, selection, calibration and final evaluation are separated chronologically, so that withheld data cannot re-enter through model selection.

The empirical work will draw on publicly available sources, in particular South African Reserve Bank published ZARONIA fixings and curve data, supplemented where available by Bloomberg terminal quotes for swaps and legacy options; data access remains the central feasibility risk, and novelty is the second.

Either an application-only contribution or a precise negative result, such as a non-identification theorem or a demonstration that no informative certificate survives realistic samples, would be a legitimate outcome.

An indicative full-time schedule runs from foundations, the novelty review and the data audit (months 1 to 6), through the transition model and identification (7 to 14) and hedging and the statistical analysis (15 to 22), to simulation and empirical evaluation (23 to 29) and synthesis and examination (30 to 36).

Month 6 is the first decision point, at which I will confirm data access, the restricted model and at least one candidate original result before expanding implementation.

The proposed contribution has two linked parts: a rigorous identification analysis for a restricted transition model and observation set, and a justified analysis of loss thresholds for

implementable hedges under specified dependence and shift. Both remain conditional on the literature audit and on successful proofs; a South African implementation is an application contribution, and geographic novelty alone will not establish the mathematical originality the degree requires.

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